Configuring App-Aware Routing SLA Classes

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Summary

This article explains how to configure app-aware routing SLA classes for NetCore SD-WAN 5.0 in NetCore Orchestrator and on NCX edge devices, then bind applications to those classes with an app-route policy. It also shows verification commands and explains how the NCS-SLA-2001 SLA Violation Detected alarm is used when measured tunnel performance exceeds the configured SLA thresholds.

Applies to: NetCore SD-WAN 5.0, NetCore Orchestrator, NCX-1000 branch edge, NCX-4000 regional hub, NCX-9000 data center edge

Overview

App-aware routing uses SLA classes to select the best compliant transport for each application. In NetCore SD-WAN 5.0, tunnels are monitored with BFD probes, and application traffic can be steered across MPLS, Internet broadband, or LTE/5G based on latency, jitter, and packet-loss measurements.

When an application path exceeds its assigned SLA class threshold for the configured breach policy, NetCore SD-WAN raises NCS-SLA-2001 SLA Violation Detected. For the full threshold reference, see KB-NET-1002. For breach timing, escalation, and automatic failover behavior, see KB-NET-1003.

NetCore SD-WAN 5.0 SLA classes

SLA class	Latency	Jitter	Loss	Typical use
Voice	<= 150 ms	<= 30 ms	<= 1%	Interactive voice applications
Video	<= 200 ms	<= 40 ms	<= 2%	Video conferencing and real-time video
Business-Critical	<= 250 ms	<= 50 ms	<= 3%	ERP, collaboration, transaction systems
Best-Effort	<= 400 ms	n/a	<= 5%	General web and non-critical applications

Prerequisites

Complete these prerequisites before configuring SLA classes or app-route policies.

- NetCore Orchestrator is running NetCore SD-WAN 5.0 and can reach the target NCX devices.
- Target edges are onboarded and show an active control connection in NetCore Orchestrator.
- At least two transports are available if you expect automatic failover behavior, for example MPLS and Internet broadband.
- BFD sessions are established between the service-side tunnel endpoints that will be used for app-aware routing.
- You have a maintenance window or approved change record for policy deployment, because applying an app-route policy can change active forwarding decisions.
- You have administrator or network-policy privileges in NetCore Orchestrator and CLI access to the target NCX edge if using the CLI workflow.

Step-by-step Procedure

Use the following procedure to define SLA classes, bind applications to the classes with an app-route policy, and deploy the policy to the required sites. The example uses a branch NCX-1000 named BR-101 and a regional hub NCX-4000 named HUB-01.

- 1. In NetCore Orchestrator, go to SD-WAN > Policies > SLA Classes and confirm that the release 5.0 defaults exist for Voice, Video, Business-Critical, and Best-Effort. If the defaults were removed, recreate them with the exact thresholds shown in the Overview table.
- 2. Create or update the app-route policy in NetCore Orchestrator. Go to SD-WAN > Policies > App-Route > Create Policy, name the policy AR-BRANCH-STANDARD, and add application rules that map voice traffic to Voice, video traffic to Video, business applications to Business-Critical, and all remaining traffic to Best-Effort.
- 3. Set the preferred transport order for each rule. A common policy is to prefer MPLS for Voice and Business-Critical, prefer Internet broadband for Video when compliant, and use Internet broadband then LTE/5G for Best-Effort.
- 4. Attach the policy to the target site group. In NetCore Orchestrator, select SD-WAN > Sites > Site Groups, open Branch-Standard, attach AR-BRANCH-STANDARD, and stage the change.
- 5. Review the staged change and deploy it. Confirm that NetCore Orchestrator reports a successful policy push to each NCX edge.
- 6. Optional CLI workflow: if your change process requires local CLI staging before Orchestrator deployment, configure the same SLA class and app-route policy values on the target edge. The CLI example below mirrors the Orchestrator policy.
- 7. After deployment, wait at least one app-route poll interval before validating statistics. In release 5.0, app-route polling occurs every 120 seconds; breach declaration requires 3 consecutive intervals, which is approximately 6 minutes. See KB-NET-1003 for full escalation behavior.

```
# Example Orchestrator policy export: AR-BRANCH-STANDARD
policy:
  name: AR-BRANCH-STANDARD
  type: app-route
  release: "5.0"
  sla_classes:
    - name: Voice
      latency_ms: 150
      jitter_ms: 30
      loss_percent: 1
    - name: Video
      latency_ms: 200
      jitter_ms: 40
      loss_percent: 2
    - name: Business-Critical
      latency_ms: 250
      jitter_ms: 50
      loss_percent: 3
    - name: Best-Effort
      latency_ms: 400
      jitter_ms: null
      loss_percent: 5
  rules:
    - sequence: 10
      match_application_group: realtime-voice
      sla_class: Voice
      transport_preference: [MPLS, Internet broadband, LTE/5G]
    - sequence: 20
      match_application_group: realtime-video
      sla_class: Video
      transport_preference: [Internet broadband, MPLS, LTE/5G]
    - sequence: 30
      match_application_group: business-critical
      sla_class: Business-Critical
      transport_preference: [MPLS, Internet broadband, LTE/5G]
    - sequence: 999
      match_application_group: any
      sla_class: Best-Effort
      transport_preference: [Internet broadband, LTE/5G, MPLS]

# Optional NCX edge CLI workflow on BR-101
ncx-edge# configure terminal
ncx-edge(config)# sdwan sla-class Voice latency 150 jitter 30 loss 1
ncx-edge(config)# sdwan sla-class Video latency 200 jitter 40 loss 2
ncx-edge(config)# sdwan sla-class Business-Critical latency 250 jitter 50 loss 3
ncx-edge(config)# sdwan sla-class Best-Effort latency 400 loss 5
```

```
ncx-edge(config)# sdwan app-route policy AR-BRANCH-STANDARD
ncx-edge(config-app-route)# rule 10 match application-group realtime-voice sla-class Voice prefer-transport
MPLS Internet-broadband LTE-5G
ncx-edge(config-app-route)# rule 20 match application-group realtime-video sla-class Video prefer-transport
Internet-broadband MPLS LTE-5G
ncx-edge(config-app-route)# rule 30 match application-group business-critical sla-class Business-Critical
prefer-transport MPLS Internet-broadband LTE-5G
ncx-edge(config-app-route)# rule 999 match application-group any sla-class Best-Effort prefer-transport
Internet-broadband LTE-5G MPLS
ncx-edge(config-app-route)# exit
ncx-edge(config)# sdwan site-policy attach app-route AR-BRANCH-STANDARD
ncx-edge(config)# commit comment "KB-NET-1001 configure app-aware routing SLA classes"
ncx-edge(config)# end
```

Verification

Verify the configuration from the edge CLI after the policy is deployed. Confirm that SLA classes are present, BFD sessions are up, control connections are stable, and app-route statistics show selected transports per SLA class.

If an SLA breach occurs after the release 5.0 breach policy is met, NetCore Orchestrator raises NCS-SLA-2001 SLA Violation Detected and automatically selects the best compliant transport. After the measured path returns within threshold, the restored condition is visible in app-route statistics; alarm handling details are covered in KB-NET-1003.

- Run show sdwan sla-class to confirm the exact SLA thresholds.
- Run show sdwan bfd sessions to confirm tunnel probes are active.
- Run show sdwan control connections to confirm the edge remains connected to NetCore Orchestrator.
- Run show sdwan app-route statistics to confirm application-to-SLA-class mapping and selected transport.
- If NCS-SLA-2001 SLA Violation Detected is active, compare the measured latency, jitter, and loss to the class threshold and review the affected transport.

```
ncx-edge# show sdwan sla-class
SLA Class             Latency(ms)  Jitter(ms)  Loss(%)  State
-------------------   -----------  ----------  -------  --------
Voice                 150          30          1        active
Video                 200          40          2        active
Business-Critical     250          50          3        active
Best-Effort           400          n/a         5        active

ncx-edge# show sdwan bfd sessions
Peer           Tunnel        Transport            State  Uptime     Last Flap
-------------  ------------  -------------------  -----  ---------  -------------------
10.40.0.11     tun1001       MPLS                 Up     12:44:18   2026-11-05 21:18:42
10.40.0.12     tun1002       Internet broadband   Up     12:43:55   2026-11-05 21:19:05
10.40.0.13     tun1003       LTE/5G               Up     11:58:10   2026-11-05 22:04:50

ncx-edge# show sdwan control connections
Orchestrator        Local Interface  State        Uptime     Last Change
------------------  ---------------  -----------  ---------  -------------------
198.51.100.20       ge0/0            Established  13:02:31   2026-11-05 21:00:29

ncx-edge# show sdwan app-route statistics
Timestamp: 2026-11-06 10:03:22 UTC
Policy: AR-BRANCH-STANDARD
Site: BR-101

App Group           SLA Class           Selected Transport     Lat(ms)  Jit(ms)  Loss(%)  Status
------------------  ------------------  ---------------------  -------  -------  -------  ---------
realtime-voice      Voice               MPLS                   42       8        0.1      compliant
realtime-video      Video               Internet broadband     86       14       0.4      compliant
business-critical   Business-Critical   MPLS                   51       10       0.2      compliant
any                 Best-Effort         Internet broadband     92       n/a      0.5      compliant

# Example breach display for NCS-SLA-2001 SLA Violation Detected
ncx-edge# show sdwan app-route statistics
Timestamp: 2026-11-06 10:11:22 UTC
Policy: AR-BRANCH-STANDARD
Site: BR-101
```

```
App Group           SLA Class  Selected Transport    Lat(ms)  Jit(ms)  Loss(%)  Status      Alarm
------------------  ---------  --------------------  -------  -------  -------  ----------  ------------------
------------------
realtime-voice      Voice      Internet broadband    74       16       0.3      compliant   -
realtime-voice      Voice      MPLS                  176      35       1.4      breached    NCS-SLA-2001 SLA
Violation Detected
```

Rollback

Rollback should restore the previous app-route policy or detach the new policy from the affected site group. Prefer rollback from NetCore Orchestrator so that the intended configuration remains consistent across the management plane and all edges.

- 1. In NetCore Orchestrator, go to SD-WAN > Policies > App-Route and open the deployment history for AR-BRANCH-STANDARD.
- 2. Select the last known-good policy version, review the application bindings and transport preferences, and click Restore Version.
- 3. Deploy the restored version to the affected site group.
- 4. If the policy must be removed entirely, detach AR-BRANCH-STANDARD from the site group and attach the previous production policy.
- 5. Use the CLI rollback example only when directed by your operations procedure or when Orchestrator policy deployment is temporarily unavailable.

```
# CLI rollback example: detach the new app-route policy and restore previous policy
ncx-edge# configure terminal
ncx-edge(config)# no sdwan site-policy attach app-route AR-BRANCH-STANDARD
ncx-edge(config)# sdwan site-policy attach app-route AR-BRANCH-PREVIOUS
ncx-edge(config)# commit comment "Rollback KB-NET-1001 app-route SLA policy"
ncx-edge(config)# end

# Verify after rollback
ncx-edge# show sdwan app-route statistics
Timestamp: 2026-11-06 10:28:44 UTC
Policy: AR-BRANCH-PREVIOUS
Site: BR-101

App Group           SLA Class           Selected Transport     Lat(ms)  Jit(ms)  Loss(%)  Status
------------------  ------------------  ---------------------  -------  -------  -------  ---------
realtime-voice      Voice               MPLS                   39       7        0.1      compliant
business-critical   Business-Critical   MPLS                   48       9        0.2      compliant
```

Known Limitations

These limitations apply to NetCore SD-WAN 5.0 app-aware routing SLA class configuration.

- Best-Effort does not evaluate jitter. Configure latency <= 400 ms and loss <= 5%; do not configure a jitter threshold for Best-Effort.
- SLA classes define path compliance only. They do not replace QoS queue configuration or bandwidth reservation. If packet drops are occurring in local queues, verify QoS separately with show policy qos-queue counters.
- NCS-SLA-2001 SLA Violation Detected is raised only after the release 5.0 breach policy is met: app-route poll interval 120 seconds with multiplier 3, or approximately 6 minutes. Do not use the superseded release 4.2 breach timing for NetCore SD-WAN 5.0 deployments.
- Automatic failover selects the best compliant transport. If no transport is compliant for the application SLA class, traffic remains on the least-violating available path until a compliant path is restored.
- Local CLI changes can be overwritten by the next NetCore Orchestrator policy deployment. Use Orchestrator as the source of truth unless your change procedure explicitly requires local staging.
- For a complete threshold matrix, see KB-NET-1002. For severity escalation, ticket creation, and auto-failover behavior, see KB-NET-1003.

Related Articles

KB-NET-1002: SLA Class Thresholds Reference (Latency, Jitter, Loss)

KB-NET-1003: SLA Breach Policy: Escalation and Auto-Failover